Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the classification of various classes using k-Nearest Neighbor approach**	
Experiment No: 03	**Date:02.02.2026**	**Enrollment No:92301733004**

Aim: To obtain the classification of various classes using k-Nearest Neighbor approach

IDE: Google Colab

Theory:

This algorithm is used to solve the classification model problems. K-nearest neighbor or K-NN algorithm basically creates an imaginary boundary to classify the data. When new data points come in, the algorithm will try to predict that to the nearest of the boundary line. Therefore, larger k value means smother curves of separation resulting in less complex models. Whereas, smaller k value tends to over fit the data and resulting in complex models. It's very important to have the right k-value when analyzing the dataset to avoid over fitting and under fitting of the dataset.

The model representation for KNN is the entire training dataset. It is as simple as that. KNN has no model other than storing the entire dataset, so there is no learning required. Efficient implementations can store the data using complex data structures like k-d trees to make look-up and matching of new patterns during prediction efficient. Because the entire training dataset is stored, you may want to think carefully about the consistency of your training data. It might be a good idea to curate it, update it often as new data becomes available and remove erroneous and outlier data.

K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories. K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm. K-NN algorithm can be used for Regression as well as for Classification but mostly it is used for the Classification problems.

K-NN is a non-parametric algorithm, which means it does not make any assumption on underlying data. It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset. KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data.

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Example: Suppose, we have an image of a creature that looks similar to cat and dog, but we want to know either it is a cat or dog. So for this identification, we can use the KNN algorithm, as it works on a similarity measure. Our KNN model will find the similar features of the new data set to the cats and dogs images and based on the most similar features it will put it in either cat or dog category.

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Pre-process the data
5. Visualize the Data
6. Separate the feature and prediction value columns
7. Select the number K of the neighbors
8. Calculate the Euclidean distance of K number of neighbors
9. Take the K nearest neighbors as per the calculated Euclidean distance.
10. Among these k neighbors, count the number of the data points in each category.
11. Assign the new data points to that category for which the number of the neighbor is maximum.
12. Our model is ready.

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Pre Lab Exercise:

a. What is K in KNN-approach?

K represents the **number of nearest neighbors** considered while classifying a new data point.

It determines how many surrounding data points influence the prediction. A smaller K can lead to overfitting, while a larger K gives smoother decision boundaries.

b. Which are the types of distance metrices?

Common distance metrics used in KNN are:

- Euclidean Distance
- Manhattan Distance
- Minkowski Distance
- Hamming Distance (for categorical data)

c. What can be the criteria of selection of the value of K?

- Choose odd values of K to avoid ties (especially in binary classification).
- Use cross-validation to find optimal K.
- Small K → sensitive to noise (overfitting).
- Large K → may ignore local patterns (underfitting).
- Typically, K ≈ √(number of data points) is a starting point.

d. What are the advantages of KNN algorithm?

- Simple and easy to implement
- No training phase (lazy learner)
- Works well with small datasets
- Can be used for both classification and regression
- No assumptions about data distribution

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Program (Code):

```
import time

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


from sklearn.datasets import load_digits

from sklearn.model_selection import train_test_split, GridSearchCV

from sklearn.preprocessing import StandardScaler

from sklearn.pipeline import Pipeline


from sklearn.neighbors import KNeighborsClassifier

from sklearn.ensemble import RandomForestClassifier

from sklearn.neural_network import MLPClassifier


from sklearn.metrics import (

    accuracy_score,

    precision_score,

    recall_score,

    f1_score,

    confusion_matrix

)
```

```
digits = load_digits()

X = digits.data

y = digits.target

images = digits.images


plt.figure(figsize=(8, 4))

for i in range(10):

    plt.subplot(2, 5, i + 1)

    plt.imshow(images[i], cmap="gray")

    plt.title(f"{y[i]}")

    plt.axis("off")

plt.suptitle("Sample Digits")

plt.tight_layout()
plt.show()


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)


knn_pipeline = Pipeline([

    ("scaler", StandardScaler()),

    ("knn", KNeighborsClassifier())

])


knn_param_grid = {
    "knn__n_neighbors": [3, 5, 7, 9, 13, 15, 20]}
```

```
rf_pipeline = Pipeline([
    ("rf", RandomForestClassifier(random_state=42))
])


rf_param_grid = {
    "rf__n_estimators": [25, 50, 100, 200, 500],
    "rf__max_depth": [None, 10, 20]}


mlp_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("mlp", MLPClassifier(
        max_iter=1000,
        random_state=42,
        early_stopping=True
    ))
])


mlp_param_grid = {
    "mlp__hidden_layer_sizes": [(64,), (128,), (64, 64)],
    "mlp__activation": ["relu", "tanh"],
    #"mlp__alpha": [0.0001, 0.001],
    "mlp__learning_rate_init": [0.001, 0.01, 1.0, 5.0, 10.0]
}


#Training the model
```

```
def train_with_timing(pipeline, param_grid):
    start = time.time()
    grid = GridSearchCV(
        pipeline,
        param_grid,
        cv=5,
        scoring="f1_weighted",
        n_jobs=-1
    )
    grid.fit(X_train, y_train)
    train_time = time.time() - start
    return grid, train_time


knn_grid, knn_train_time = train_with_timing(knn_pipeline, knn_param_grid)
rf_grid, rf_train_time = train_with_timing(rf_pipeline, rf_param_grid)
mlp_grid, mlp_train_time = train_with_timing(mlp_pipeline, mlp_param_grid)


#Inferencing the model
def inference_time(model):
    start = time.time()
    model.predict(X_test)
    return time.time() - start


knn_test_time = inference_time(knn_grid.best_estimator_)
rf_test_time = inference_time(rf_grid.best_estimator_)
```

```
mlp_test_time = inference_time(mlp_grid.best_estimator_)


#Evaluation

def evaluate_model(model):

    y_pred = model.predict(X_test)

    return {

        "Accuracy": accuracy_score(y_test, y_pred),

        "Precision": precision_score(y_test, y_pred, average="weighted"),

        "Recall": recall_score(y_test, y_pred, average="weighted"),

        "F1-score": f1_score(y_test, y_pred, average="weighted")

    }


knn_metrics = evaluate_model(knn_grid.best_estimator_)

rf_metrics = evaluate_model(rf_grid.best_estimator_)

mlp_metrics = evaluate_model(mlp_grid.best_estimator_)


#Result with comparison

results_df = pd.DataFrame(

    [knn_metrics, rf_metrics, mlp_metrics],

    index=["KNN", "Random Forest", "MLP"]

)


results_df["Training Time (s)"] = [

    knn_train_time, rf_train_time, mlp_train_time

]
```

```
results_df["Inference Time (s)"] = [
    knn_test_time, rf_test_time, mlp_test_time
]

print(results_df)

results_df.iloc[:, :4].plot(kind="bar", figsize=(10, 6))
plt.title("Model Performance Comparison (Digits)")
plt.ylabel("Score")
plt.ylim(0, 1)
plt.xticks(rotation=0)
plt.grid(axis="y")
plt.show()

models = {
    "KNN": knn_grid.best_estimator_,
    "Random Forest": rf_grid.best_estimator_,
    "MLP": mlp_grid.best_estimator_
}

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, (name, model) in zip(axes, models.items()):
    cm = confusion_matrix(y_test, model.predict(X_test))
```

```
    sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", ax=ax)

    ax.set_title(name)

    ax.set_xlabel("Predicted")

    ax.set_ylabel("Actual")


plt.tight_layout()
plt.show()
```

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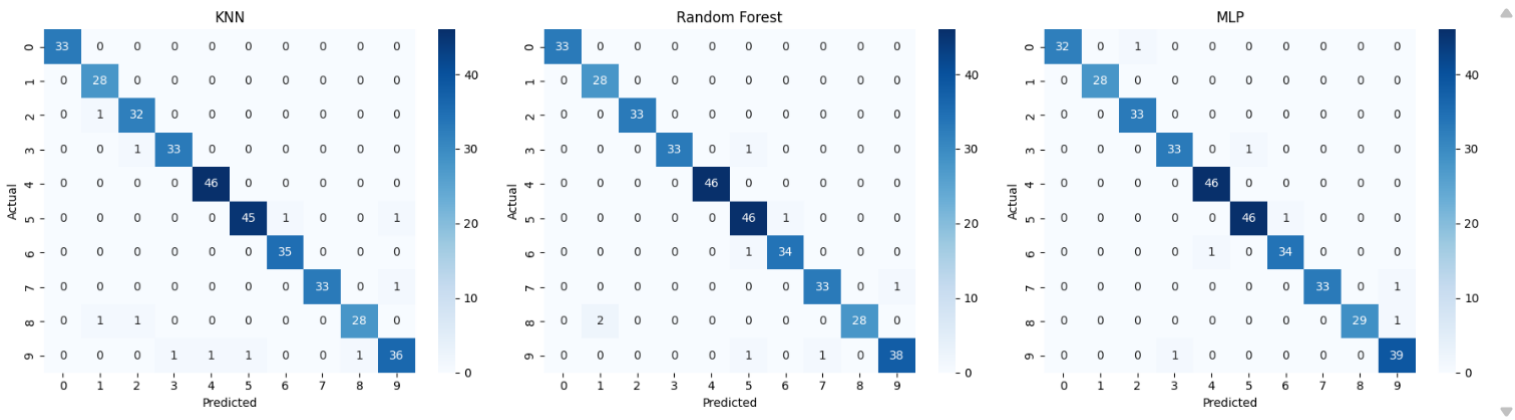
Results:

To be attached with

a. Confusion matrix



b. Accuracy Score

	Accuracy	Precision	Recall	F1-score	Training Time (s)	\
KNN	0.969444	0.969666	0.969444	0.969287	5.000380	
Random Forest	0.977778	0.978417	0.977778	0.977820	49.642176	
MLP	0.980556	0.980832	0.980556	0.980577	58.312733	

	Inference Time (s)
KNN	0.019512
Random Forest	0.106671
MLP	0.003206

Observation and Result Analysis:

a. **Nature of the dataset**

The dataset consists of labeled data with multiple features used for classification. It may include numerical values and requires preprocessing like normalization for better performance.

b. **During Training Process**

KNN does not perform actual training. It simply stores the entire dataset. No model is built, making it a lazy learning algorithm.

c. **After the training Process**

When a new data point is given, the algorithm calculates distances from all training points, selects K nearest neighbors, and assigns the class based on majority voting.

d. **Observation over the Learning Curve**

- Small K → high variance, low bias (overfitting)
- Large K → low variance, high bias (underfitting)
- Optimal K provides a balance between bias and variance

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Post Lab Exercise:

a. **Is KNN useful in regression-based problems?**

Yes, KNN can be used for regression by taking the **average of K nearest neighbors' values** instead of majority voting.

b. **Why KNN is a non-parametric algorithm?**

Because it **does not assume any underlying data distribution** and does not learn parameters from data.

c. **What are the assumptions of KNN approach?**

- Similar data points exist close to each other
- Data is properly scaled
- Distance metric correctly reflects similarity

d. **How can the KNN approach make the prediction of the unseen dataset?**

It calculates the distance between the new data point and all training points, selects the K nearest ones, and assigns the class based on majority voting.

e. **Why is KNN called a Lazy Learner?**

Because it **does not learn during training** and only performs computation during prediction time.

f**. Why should we not use KNN for large dataset?**

- High computation time (distance calculation for all points)
- High memory usage (stores entire dataset)
- Slow prediction performance